

What is this?

This notebook is part of a collection of `pluto` notebooks on various topics discussed during the Time Domain Astrophysics course delivered by Stefano Covino at the Università dell'Insubria in Como (Italy). Please direct questions and suggestions to stefano.covino@inaf.it.

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TIME-DOMAIN ASTROPHYSICS

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Statistics Reminder

LIKELIHOOD

The probability of "B" being True, given "A" is True

PRIOR

The probability "A" being True. This is the knowledge.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

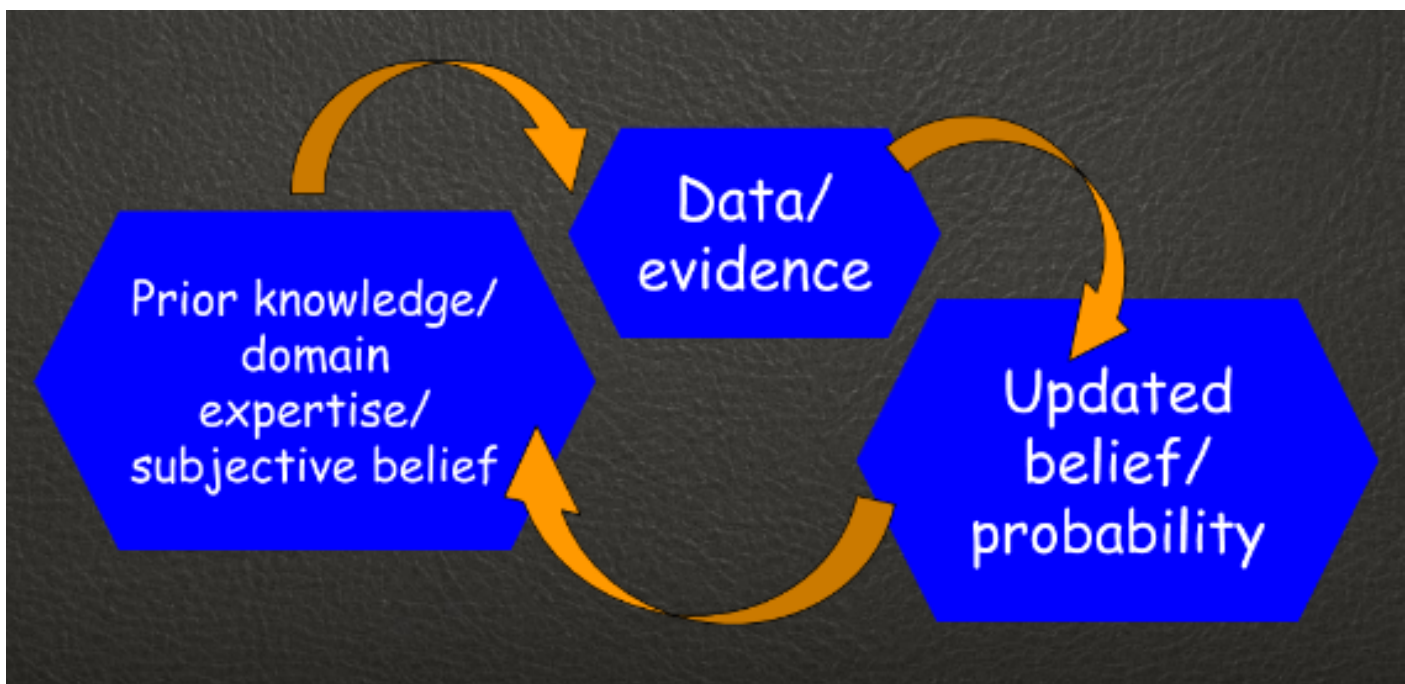
POSTERIOR

The probability of "A" being True, given "B" is True

MARGINALIZATION

The probability "B" being True.

- Bayes' theorem brings in the concept of "degree of belief" into hard statistical modelings. Bayes' rule is the only mechanism that can be used to gradually update the probability of an event as the evidence or data is gathered sequentially.
- There is an intriguing line of thought that describes the law of probability as the natural scenario to formalize the logic of science. See, for instance, [Cox's theorem](#).



Bayes' theorem: example of application

- Given a disease, D , with a *prevalence* 10%, and a test, T , with a *sensitivity* 95% and *specificity* 85%, what is the probability that subjects who test positive for D really is affected by D ?

- Data:
 - $p(D=1) = 0.1$, 1.5 *prevalence*
 - $p(T=1 \mid D=1) = 0.95$ *sensitivity*
 - $p(T=0 \mid D=0) = 0.85$ *specificity*

$$p(D = 1|T = 1) = \frac{p(T = 1|D = 1)p(D = 1)}{p(T = 1|D = 1)p(D = 1) + p(T = 1|D = 0)p(D = 0)} =$$

$$= \frac{0.95 \times 0.1}{0.95 \times 0.1 + 0.15 \times 0.9} = 0.413$$

- Indeed, a test of modest reliability.

Prevalence:

Sensitivity:

Specificity:

Test: 0.41

Information criteria and hypothesis testing

- In a frequentist scenario hypothesis testing is often carried out comparing the likelihood ratio of different models or the residuals by, e.g. a F-test.
 - These tests are either related to the assumptions that data are drawn from a Normal parental distributions or do not take into account the number of parameters of the models. There are more limitations we don't discuss here (see, e.g., Protassov et al. 2002).
- In a Bayesian scenario everything we can know given the data and the prior information is stated by the Bayes' Theorem:

$$p(M, \theta | D, I) = \frac{p(D | M, \theta, I) p(M, \theta | I)}{p(D | I)}$$

- The posterior probability for model M can be computed by the posterior distribution of the parameters after having marginalized (i.e. integrated) over the parameter space.
- Assuming to have two (or more) competing models, computing the *evidence*, the denominator of the previous expression, we can evaluate the **odds ratio**:

$$O_{21} \equiv \frac{p(M_2 | D, I)}{p(M_1 | D, I)} = \frac{p(M_2 | I)}{p(M_1 | I)} \frac{p(D | M_2)}{p(D | M_1)} = \frac{p(M_2 | I)}{p(M_1 | I)} B_{ij}$$

- The first factor of the odds ratio is the prior odd ratio, and the second factor is called the **Bayes factor**.
- This is a very sophisticated approach but, often, computationally hard to apply. There are simpler tools, although with limitations and assumptions.
- Let's follow a simple example, just to understand how a Bayes factor works: we are comparing two models: M_1 , with a single parameter. θ , and M_0 with θ fixed at a given value θ_0 . So, M_0 has no free parameters.
- In most cases the likelihood function, $\mathcal{L}(\theta) = p(D | \theta, M_1)$, is much more peaked than the prior, so that the prior is essentially constant over the range of significant likelihood. Therefore, if $\hat{\theta}$ is the maximum of likelihood, we can write:

$$p(D | M_1) \sim p(\hat{\theta} | M_1) \int \mathcal{L}(\theta) d\theta$$

- Then, we write the integral of the likelihood as the product of its maximum times a characteristic width (e.g., a full width at half maximum, etc.):

$$p(D | M_1) \sim p(\hat{\theta} | M_1) \mathcal{L}(\hat{\theta}) \delta\theta$$

- And finally, for clarity, we write $p(\hat{\theta} | M_1)$ as the reciprocal of a characteristic width of the prior, $\Delta\theta$. Thus:

$$p(D | M_1) \sim \mathcal{L}(\hat{\theta}) \frac{\delta\theta}{\Delta\theta}$$

- Since M_0 has no free parameters $p(D | M_0) = \mathcal{L}(\theta_0)$, and the Bayes factor turns out to be:

$$B_{10} \sim \frac{\mathcal{L}(\hat{\theta})}{\mathcal{L}(\theta_0)} \frac{\delta\theta}{\Delta\theta}$$

- The likelihood ratio can never be less than unity, since θ_0 is within the possible range of θ , and the second factor penalizes the likelihood ratio for the "wasted" parameter space that gets ruled out by the data ($\delta\theta$ is narrower than $\Delta\theta$).

In any case, a full discussion of hypothesis testing in a Bayesian scenario is a very interesting topic, but definitely beyond the purpose of the course. Nevertheless, during the course, we will often mention this matter.

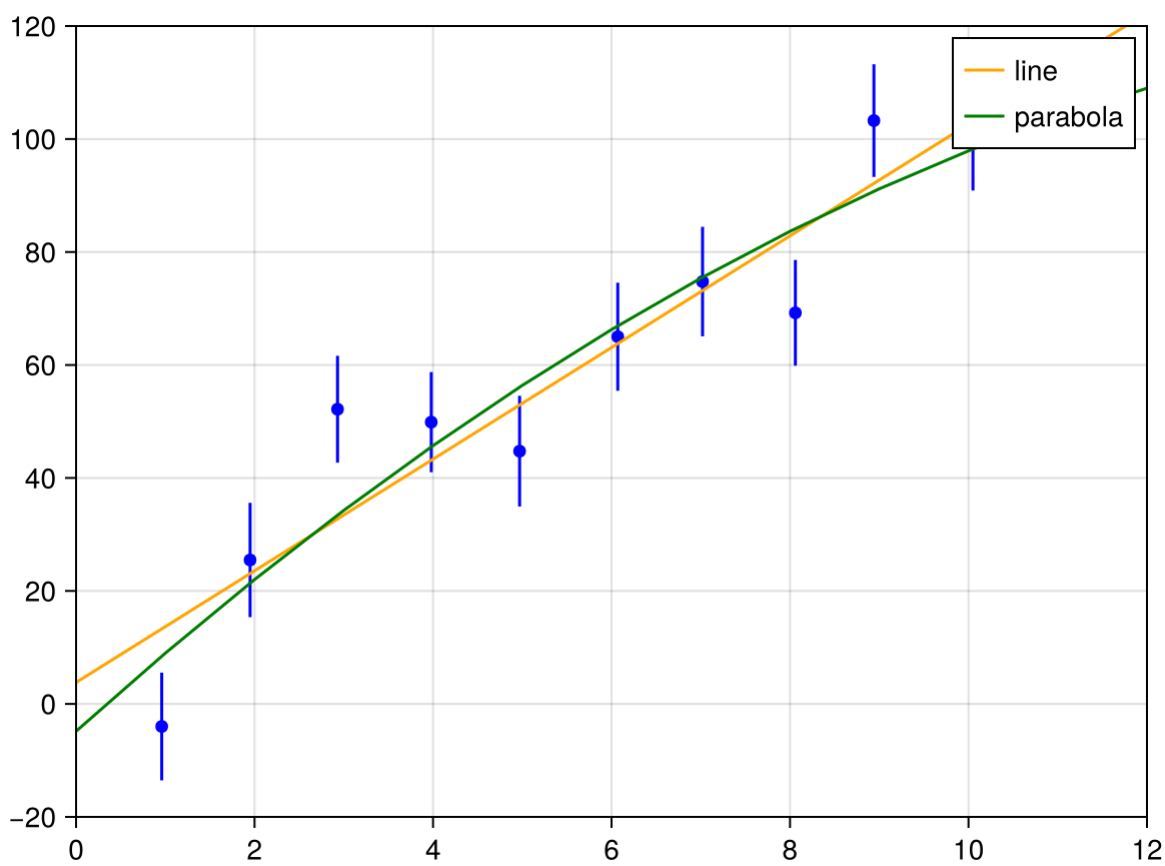
- There are simpler alternatives, although with further assumptions and limitations.
 - However, they require to compute only the maximum likelihood, and not the whole posterior distribution of parameters.
- A well known criterion is the Bayesian (or Schwarz) Information Criterion (BIC):

$$BIC \equiv -2 \log[L^0(M)] + k \log N$$

- where $L^0[M]$ is the maximum likelihood for model M , k the number of model parameters and N is the number of data points.
- Another frequently applied criterion is the Akaike Information Criterion (AIC):

$$AIC \equiv -2 \log[L^0(M)] + 2k + \frac{2k(k+1)}{N-k-1}$$

- When multiple models are compared, the one with the smallest AIC or BIC is the best model to select. If the models are equally successful in describing the data (they have the same value of $L^0[M]$), then the model with fewer free parameters "wins".



```

1 begin
2   # Let's define two models for a given dataset
3
4   f1(x;a=1,b=0) = a.*x .+ b
5   f2(x;c=1,a=1,b=0) = c.*x.^2 + a.*x .+ b
6
7   χ21(prs) = sum((f1(x,a=prs[1],b=prs[2]) .- y).^2 ./ σ.^2)
8   χ22(prs) = sum((f2(x,c=prs[1],a=prs[2],b=prs[3]) .- y).^2 ./ σ.^2)
9
10
11  fg1 = Figure()
12
13
14  # data
15  x = [0.96,1.95,2.93,3.98,4.97,6.07,7.02,8.06,8.94,10.05]
16  y = [-3.99,25.48,52.17,49.88,44.745,65.00,74.76,69.23,103.26,100.37]
17  σ = [9.53,10.12,9.46,8.86,9.80,9.56,9.69,9.35,9.97,9.48]
18
19
20  x01 = [10.0,5.0]
21  res1 = optimize(χ21, x01)
22
23  x02 = [1.,10.0,5.0]
24  res2 = optimize(χ22, x02)
25
26
27  prs1 = Optim.minimizer(res1)
28  prs2 = Optim.minimizer(res2)
29
30  #printfmtln("ML for model 1 = {:.2f} and model 2 = {:.2f}",χ21(prs1),χ22(prs2))
31
32  bc1 = BIC(χ21(prs1),length(x),2)
33  bc2 = BIC(χ22(prs2),length(x),3)
34
35  #printfmtln("BIC for model 1 = {:.2f} and model 2 = {:.2f}",bc1,bc2)
36
37

```

```

38     ax1fg1 = Axis(fg1[1, 1])
39
40
41     scatter!(x,y,color=:blue)
42     errorbars!(x,y,σ,color=:blue)
43     lines!(0:12,f1(0:12,a=prs1[1],b=prs1[2]),color=:orange,label="line")
44     lines!(0:12,f2(0:12,c=prs2[1],a=prs2[2],b=prs2[3]),color=:green,label="parabola")
45
46     xlims!(0,12)
47     ylims!(-20,120)
48
49     axislegend()
50
51     fg1
52
53
54 end

```

ML for model 1 = 12.51 and model 2 = 11.59

BIC for model 1 = −20.42 and model 2 = −16.27

- In spite of the decrease of the residuals the simpler fit is considerably preferred $\Delta BIC \sim 4$ (let's remember it is a logarithmic scale).
 - It is also possible to show that, under several hypotheses, $\log \textit{Evidence} \approx -BIC/2$.
 - If we use decibel units, the support for the simpler model turns out to be ~ 18 dB.
-
- Let's try an example of Bayes' theorem application from astronomy:

Computing the distance of an open cluster

- We infer the distance to a star from a parallax measurement. The true but unknown distance d in kiloparsecs (Kpc) is related to the true but unknown parallax π in milliarcseconds (mas) through:

$$d[kpc] = \frac{1}{\pi} [mas]$$

-
- We are going to use parallax measurements with some fixed uncertainty, σ , that we treat as known.
- In the Bayesian framework, we wish to infer the parameter d given the data drawing the posterior distribution of our parameters.

$$p(d|y) \sim p(y|d)p(d)$$

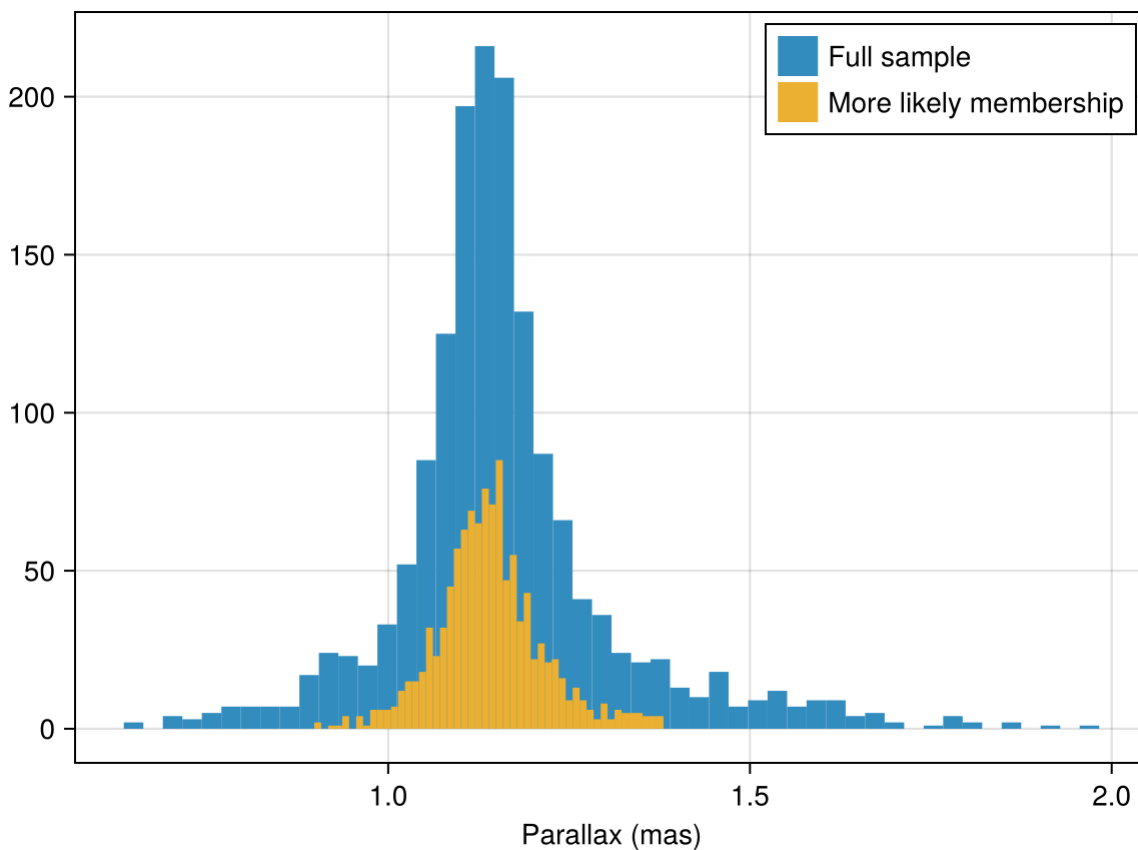
Data

- Let's load in the data for the NGC2682 (M67) open star cluster

```
1 begin
2   hdu = FITS("NGC_2682.fits")
3
4
5   # extract membership probabilities from kinematics (Gaia DR2)
6   pmem = read(hdu[2], "HDBscan_MemProb")
7
8   # extract parallaxes (Gaia DR2)
9   # note: no corrections have been applied for any systematics
10  p = read(hdu[2], "Parallax")
11  pe = read(hdu[2], "Parallax_Err")
12
13  # And select only objects with some probability to be part of the cluster.
14  sel = pmem .> 0.3
15 end;
```

```
1 begin
2    $\mu p$ ,  $\sigma p$  = WeightedArithmeticMean(p[sel], pe[sel])
3
4   #printfmtln("Parallax weighted mean: {:.3f} mas and error on the mean: {:.3f} mas",
5    $\mu p$ ,  $\sigma p$ )
6 end;
```

Parallax weighted mean: 1.133 mas and error on the mean: 0.001 mas



Field star parallax measurements

- Before considering the (actually simpler) case of parallaxes for stars in a cluster, let's see how a Bayesian analysis for measurement of field stars (i.e. not in a cluster) could be carried out.

The likelihood

- we may adopt a regular Gaussian likelihood for the measured distances, i.e. $\mathcal{L} \sim N(1/d, \sigma^2)$, i.e.:

$$p(y|d) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y-1/d)^2}{2\sigma^2}}$$

The Priors

- As for any Bayesian analysis, we have to define the prior information we have. And we are going to realize that this is a far from trivial problem.
- We have three different possible choices, although only the last is describing correctly our problem:
 - A uniform distribution for the parallax.
 - A uniform distribution for the distance.
 - A physically motivated prior for the space distributon of stars we are studying.
- While the prior for distance above may appear non-informative (in a sense that it is uniform), it actually encodes a strong assumption about the number density of stars ρ as a function of distance.
 - The prior implies that we are just as likely to observe stars at large distances as we are at smaller distances.
 - However, as we look out into space, the area of the solid angle defined by the distance d increases, and this in turn implies that the stellar number density is decreasing with

distance.

- Bailer-Jones et al. (2018) introduced a better prior for the parallax inference problem.
- The physical volume dV probed by an infinitesimal solid angle on the sky $d\Omega$ at a given distance d scales as the size of a shell so that $d\Omega \propto d^2$. This means that, assuming a constant stellar number density ρ everywhere, a prior behaving as $p(d) \propto d^2$ is more appropriate.
- However, we know that our Sun sits in the disk of the Galaxy, and that the actual stellar density ρ as we go radially outward in the disk should decrease as a function of distance.
- Assuming we are looking outward, and that the stellar density decreases exponentially with a length scale L (so that for a given distance we have $p(\rho|d) \propto e^{-d/L}$) the prior on distance is:

$$p(d) \propto \begin{cases} d^2 e^{-d/L} & d_{\min} < d < d_{\max} \\ 0 & \text{otherwise} \end{cases}$$

- which is the density function of a truncated Gamma(3, L) distribution.

A uniform distribution in parallax is not a uniform distrbution in distance, $d = 1/\pi$. Actually, if π is uniformly distributed between a and b , the inverse is distributed as $1/d^2$ between b^{-1} and a^{-1} .

It is clear how uniform priors are usually far from being uninformative.

Posterior distribution

- Once the prior distribution(s) and the likelihood function are specified, the posterior distribution is uniquely determined.
 - Often, the denominator quantity $p(\mathbf{y})$ is not available analytically and can be estimated by numerical integration.
- The posterior distribution enables inference of model parameters or of quantities that can be derived from model parameters.
 - For example, the posterior mean $\mathcal{E}(\boldsymbol{\theta}|\mathbf{y})$ is a point summary for $\boldsymbol{\theta}$. The posterior distribution can also be used to define credible intervals for parameters that provide a range of probable values.

We stress that credible intervals are not confidence intervals; a 95% credible interval suggests that there is a 95% probability that the parameter lies within the specified range given our prior beliefs, the model, and the data, whereas a 95% confidence interval suggests that if similar intervals are properly constructed for multiple datasets then 95% of them are expected to contain the true (fixed) parameter value.

- The posterior distribution also provides a useful way to obtain estimates and credible intervals for other quantities of physical interest.
 - For example, if a model has parameters $\boldsymbol{\theta} = (\alpha, \beta, \kappa)$ and there is some physical quantity described by e.g., $\gamma = \beta^2/\alpha e^\kappa$, then for every sample of $\boldsymbol{\theta}$, a sample of γ can be calculated. Thus, a distribution of the physically interesting quantity γ is obtained, which can also be used to obtain point estimates and credible intervals.
- Here, we are interested in inferring the parameter for the distance $d = 1/\pi$ given the measured parallax y and its known associated measurement uncertainty σ .
- Bayes' theorem, the posterior is:

$$p(d|\mathbf{y}) \sim p(\mathbf{y}|d)p(d)$$

For the three possible priors discussed previously, and recalling that if f is uniformly distributed the PDF of the reciprocal, $y = 1/f$, is $\sim \frac{1}{y^2}$, we have:

- Uniform parallax:

$$p(d|\mathbf{y}) \sim \frac{1}{d^2} e^{-\frac{(y-1/d)^2}{2\sigma^2}}.$$

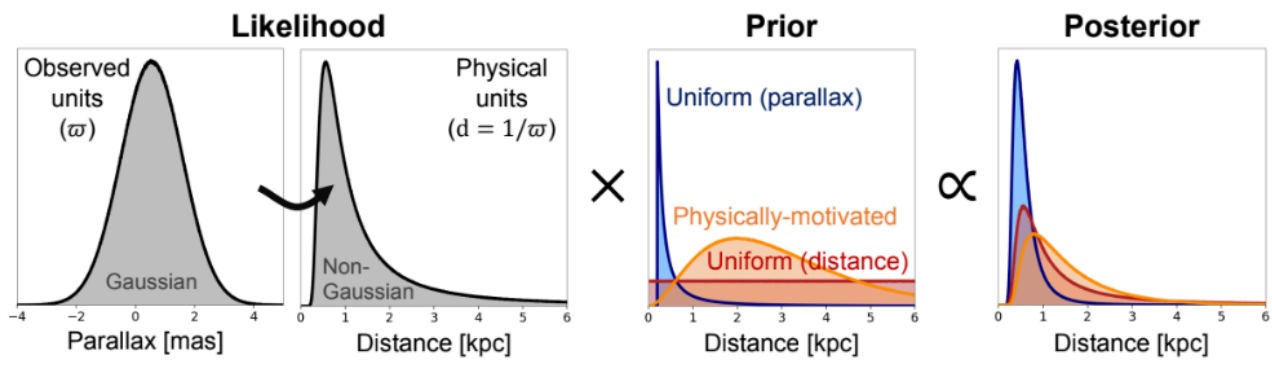
- Uniform distance:

$$p(d|\mathbf{y}) \sim e^{-\frac{(y-1/d)^2}{2\sigma^2}}.$$

- Bailer-Jones:

$$p(d|\mathbf{y}) \sim d^2 e^{-\frac{d}{L} - \frac{(y-1/d)^2}{2\sigma^2}}.$$

- for $d_{\min} < d < d_{\max}$ (and 0 otherwise).



Cluster (M67) distance

.....

- What we have written so far holds for a single measurement. However, for the M67 distance, we work with a set of n measurements for stars that we might now think are located at approximately the same distance d_{cluster} with measured parallaxes $y = y_1, y_2, \dots, y_n$, that we consider independent given d_{cluster} .
- For each measurement the Gaussian likelihood can still be applied, and the combined likelihood is the product of the individual likelihoods, again assuming the measurement uncertainties, σ_i , be known:

$$p(y_1, y_2, \dots, y_n | 1/d_{\text{cluster}}) = \prod_{i=1}^n p(y_i | 1/d_{\text{cluster}})$$

- The prior information about the distance of the cluster (from past measurements, etc.) is now a simple Normal distribution, e.g. $p(d_{\text{cluster}}) \sim \mathcal{N}(\mu = 1.2, \sigma = 0.1)$.
- Finally, the full posterior turns out to be:

$$p(d_{\text{cluster}} | y) \sim p(d_{\text{cluster}}) \prod_{i=1}^n p(y_i | 1/d_{\text{cluster}})$$

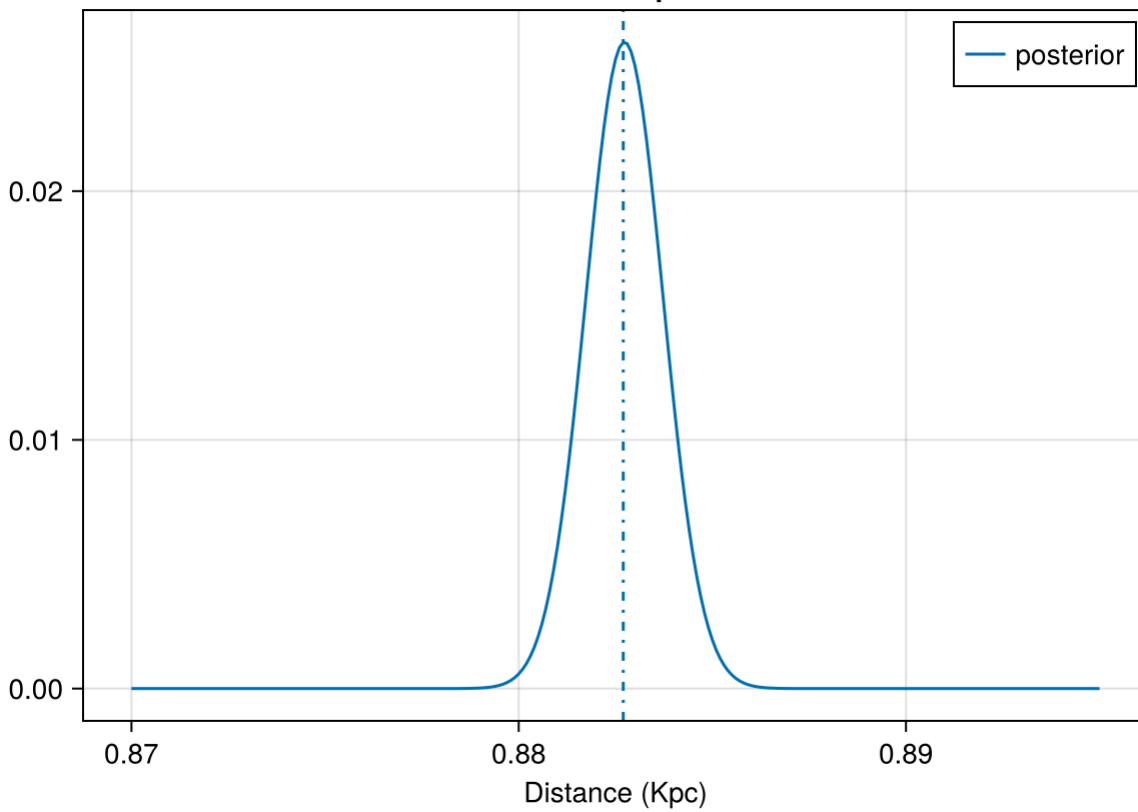
- Now, since the prior PDF do not (of course) depend on the measurements the only factor affected by the product is $\prod_{i=1}^n \exp(-\frac{(y_i - 1/d_{\text{cluster}})^2}{2\sigma_i^2})$, and since the product of n independent Gaussian densities with known variances is a Gaussian with precision parameter $\tau^{-2} = \sum_{i=1}^n \sigma_i^{-2}$ and mean parameter $1/d_{\text{cluster}}$, defining, $y_{\text{eff}} = \tau^2 \sum_{i=1}^n y_i / \sigma_i^2$ we can write:

$$p(d_{\text{cluster}} | y) \sim p(d_{\text{cluster}}) e^{-\frac{(y_{\text{eff}} - 1/d_{\text{cluster}})^2}{2\tau^2}}$$

- As mentioned above, please note that a prior over $p(d_{\text{cluster}})$, which governs the distribution of clusters of stars, is not the same as a prior over $p(d)$, which governs the distribution of individual stars.
- The posterior PDF of the cluster distance is nicely consistent with the inverse of the simple average of the star's cluster parallaxes (dot-dashed line). This happens since data are abundant and of high quality and supersede the prior information making them of low relevance.

This simple example also clarifies why, even if a Bayesian approach is far superior, frequentist results can often be adequate.

Posterior plot



- The posterior PDF of the cluster distance is nicely consistent with the inverse of the simple average of the star's cluster parallaxes (dot-dashed line). This happens since data are abundant and of high quality and supersede prior information making them of low relevance.

This simple example also clarifies why, even if a frequentist result is often enough, the Bayesian approach is far superior providing the full PDF of the parameter(s) of interest given the data.

Analysis of the posterior distribution of parameters

- The previous example is one-dimensional and involving only Gaussians can be studied analytically. However, as soon as the number of parameters grows, direct exploration of the posterior PDF becomes impractical.
- One needs sampling tools. The most widely used is the Markov Chain Monte Carlo (MCMC), yet there are several alternatives.
- Having produced a posterior $p(\theta)$, in general we might want to estimate integrals as:

$$I(\theta) = \int g(\theta)p(\theta)d\theta$$

- e.g. for marginalization $g(\theta) = \mathbf{1}$, average $g(\theta) = \theta$, credible intervals, etc.

Markov Chain Monte Carlo

- When a direct evaluation of an integral, numerically or analytically, is impossible a Monte Carlo approach is often feasible.
- Let's generate M values of the parameter set uniformly sampled within the integration volume V_θ . The integral turns out to be:

$$I \approx \frac{V_\theta}{M} \sum_{j=1}^M g(\theta_j) p(\theta_j)$$

- The algorithm works, but it is very inefficient in particular for high-dimensional integral.
- MCMC methods return a sample of points, or chain, from the k-dimensional parameter space, with a distribution that is asymptotically proportional to $p(\theta)$. With such a chain the integral becomes:

$$I \approx \frac{1}{M} \sum_{j=1}^M g(\theta_j)$$

- E.g., to estimate the expectation value for θ (i.e., $g(\theta) = \theta$), we simply take the mean value of all θ in the chain.

A Markov chain is a sequence of random variables where a given value depends only on its preceding value. Given the present value, past and future values are independent. The process generating such a chain is called the Markov process.

- A "daily life" (irony here) example of Markov chain could be a model of a baby's behavior, you might include "playing," "eating," "sleeping," and "crying" as states, which together with other behaviors could form a 'state space': a list of all possible states.
- In addition, on top of the state space, a Markov chain tells you the probability of hopping, or "transitioning," from one state to any other state, e.g., the chance that a baby currently playing will fall asleep in the next five minutes without crying first, etc.
- Coming back to "serious stuff", the process generating such a chain is called the Markov process and can be described as:

$$p(\theta_{i+1}|\{\theta_i\}) = p(\theta_{i+1}|\theta_i)$$

- i.e., it depends only on the previous step (and it is called "of the first order").

- To reach an equilibrium, or stationary, distribution of positions, it is necessary that the transition probability be symmetric:

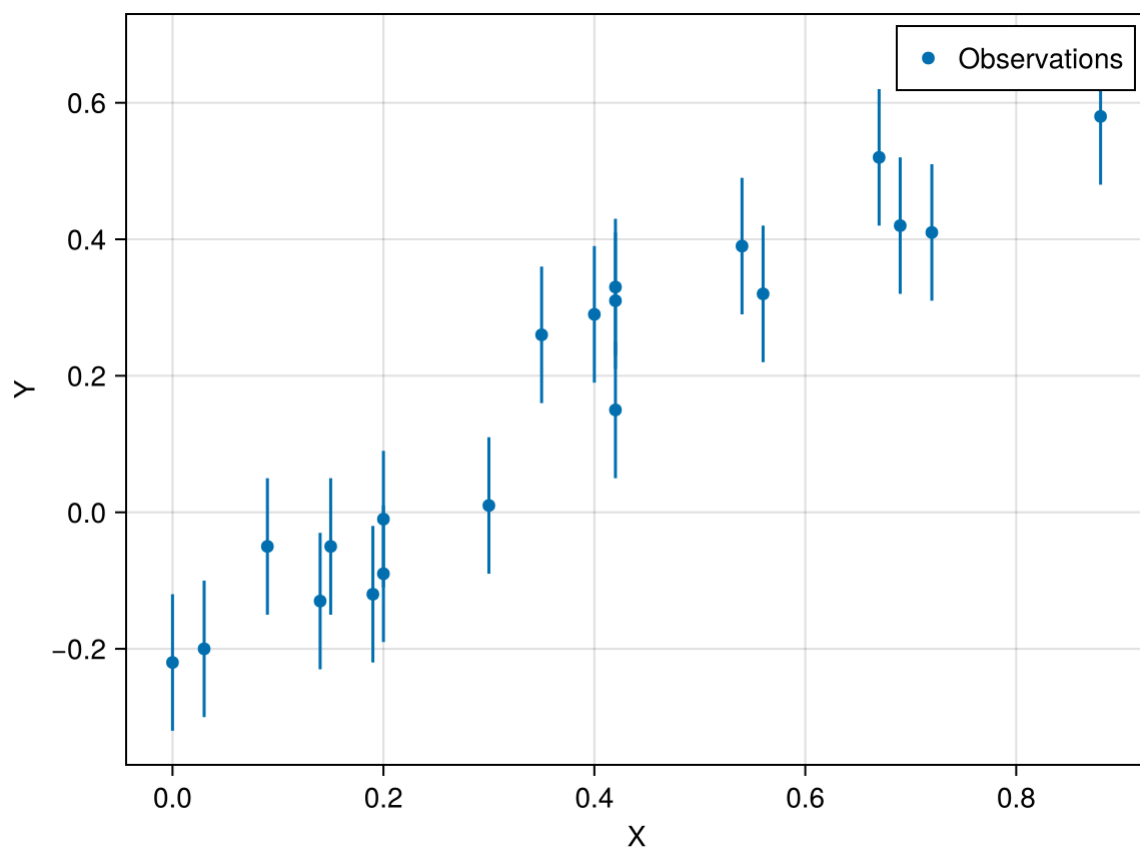
$$p(\theta_{i+1}|\theta_i) = p(\theta_i|\theta_{i+1})$$

- There are many algorithms for producing Markov chains that reach some prescribed equilibrium distribution, $p(\theta)$, and this is a field of very active research.

Interactive demo: <https://chi-feng.github.io/mcmc-demo/>

A probabilistic model

- Let's now write a simple probabilistic model to fit a straight line.



	iteration	chain	theta	q_perp	lp	n_steps	is_accept	acceptance_rate	al
1	1001	1	0.796666	-0.136565	17.0113	7.0	1.0	0.712466	
2	1002	1	0.761371	-0.12307	17.1651	11.0	1.0	0.999162	
3	1003	1	0.696572	-0.107368	16.5976	5.0	1.0	0.941679	
4	1004	1	0.822608	-0.171359	17.041	7.0	1.0	0.938458	
5	1005	1	0.667105	-0.098093	15.743	5.0	1.0	0.890161	
6	1006	1	0.681283	-0.0970874	15.9852	3.0	1.0	0.987786	
7	1007	1	0.716038	-0.107172	16.7063	3.0	1.0	0.955232	
8	1008	1	0.641146	-0.0874225	14.7396	3.0	1.0	0.867034	
9	1009	1	0.686468	-0.104765	16.3498	3.0	1.0	1.0	
10	1010	1	0.714014	-0.110518	16.8807	3.0	1.0	0.826481	
	altro								

```

1 begin
2   # Declare our Turing model
3   @model function jmodel(x,ey,y)
4     # Our prior belief
5     theta ~ Uniform(-0.5 *  $\pi$ , 0.5 *  $\pi$ )
6     q_perp ~ Normal(0.,5.)
7     #
8     m = tan.(theta)
9     q = q_perp./cos.(theta)
10    #
11    #for i in 1:length(y)
12    #  y[i] ~ Normal.(m*x .+ q,ey)
13    #end
14    return y ~ MvNormal(m*x .+ q,ey)
15  end
16
17  # Please, pay attention how we write the parameters of the models.
18  # This allows us to use simple priors!
19
20  # Let's sample our model with 2000 iterations, 1000 of them as burn-in,
21  #and 4 different chains.
22
23  iterations = 2000
24  burnins = 1000
25  chains = 4
26
27  # We apply a No-U-Turn sampler
28
29  chain = mapreduce(c -> sample(jmodel(x2,ey2,y2), NUTS(burnins,0.65), iterations),
30    chainscat, 1:chains)
31 end

```

Sampling

100%

Found initial step size

ϵ : 0.025

Sampling

100%

Found initial step size

ϵ : 0.0125

Sampling

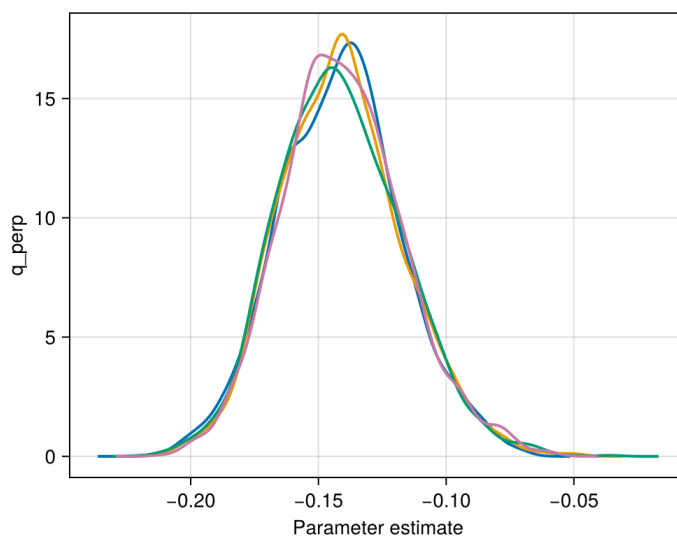
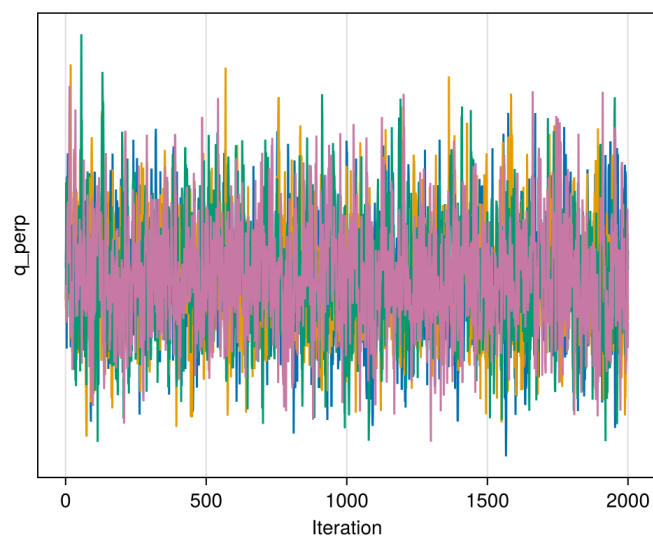
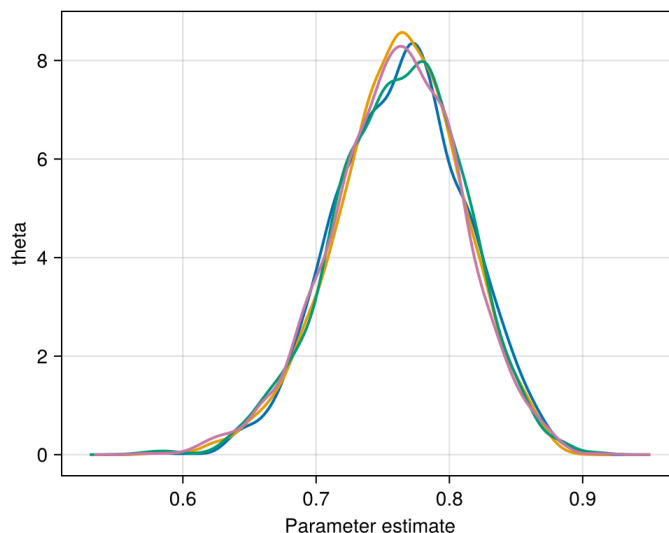
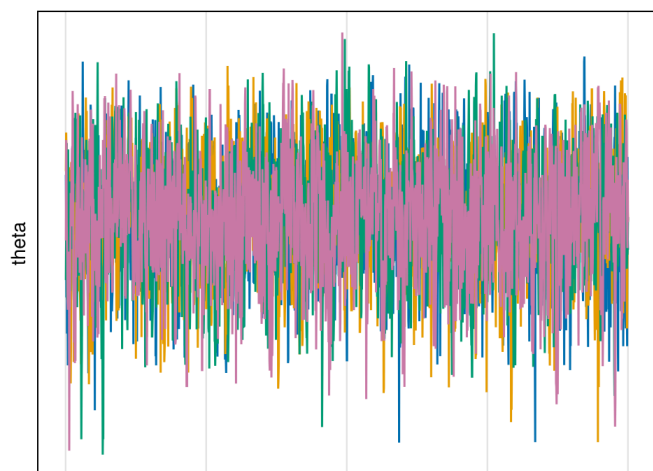
Found initial step size

ϵ : 0.025

Sampling

Found initial step size

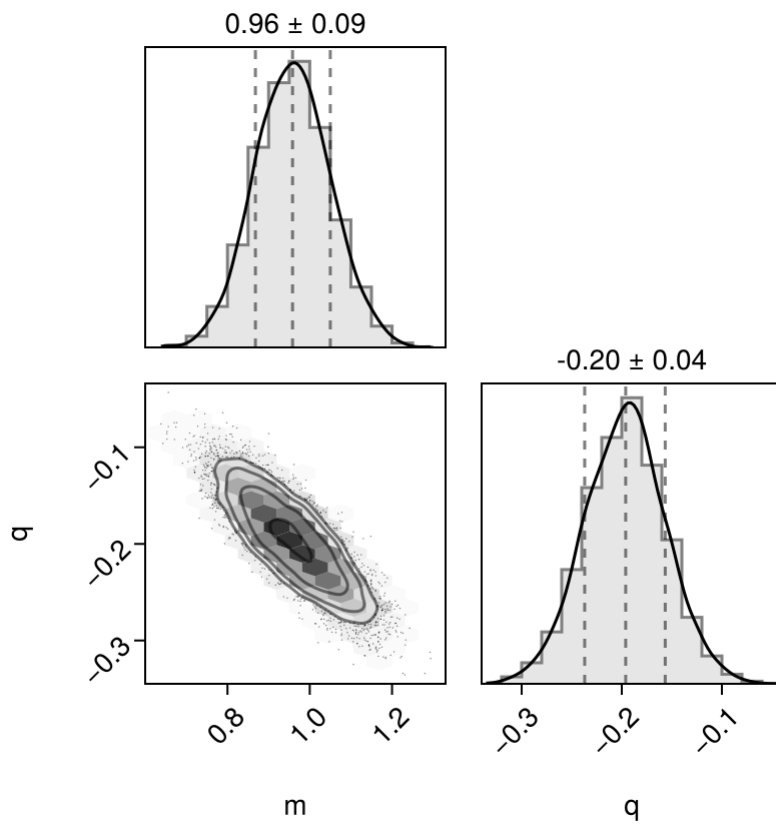
ϵ : 0.025



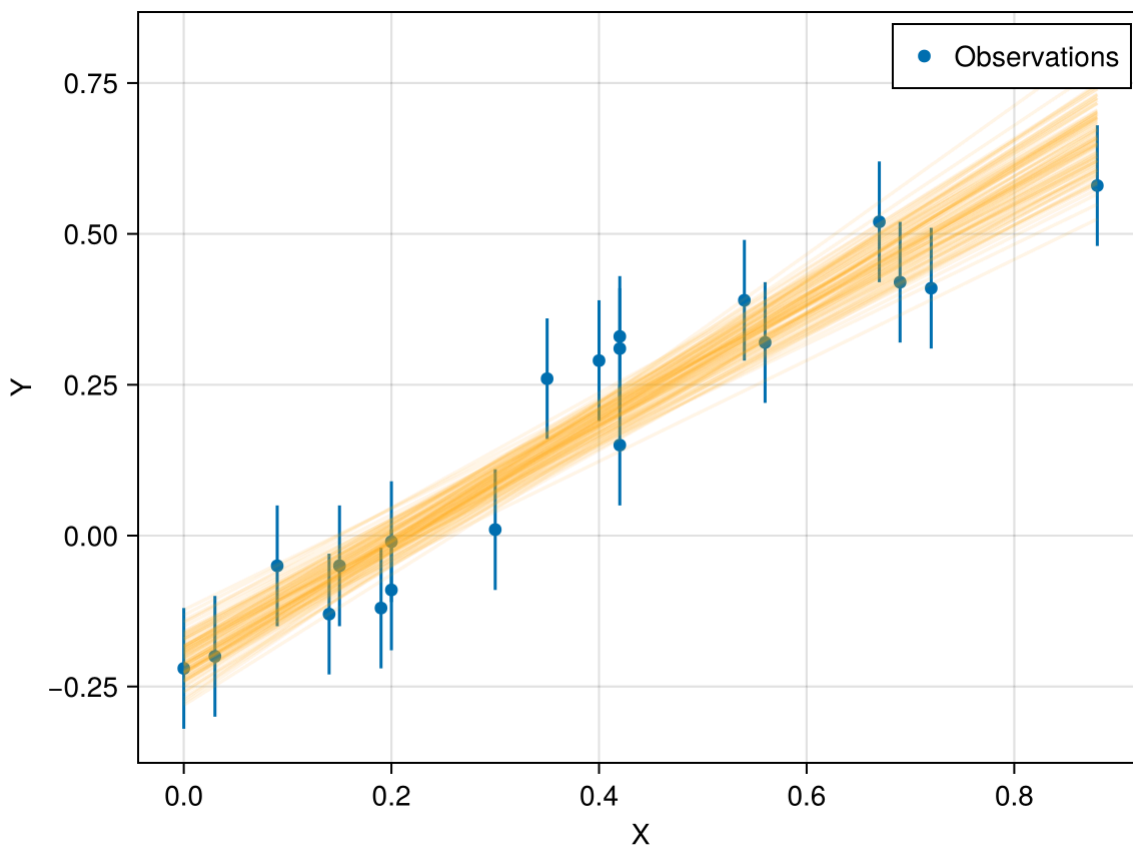
- Checking chain convergence is very important.
- The $\hat{r}$ parameter, or Potential Scale Reduction Factor (PSRF), is a MCMC convergence diagnostic that compares between-chain variance to within-chain variance for multiple chains.
- It measures if chains have mixed well; a value of $\hat{r} \approx 1$ indicates convergence.
- It has been introduced in [Gelman & Rubin \(1992\)](#).

	parameters	rhat
1	:theta	1.00057
2	:q_perp	1.00087

- The chains look fine.
- Let's produce a *corner plot*:



- And let's plot a random selection of solutions from the posterior:



- As we know, in classical statistics, models are evaluated by hypothesis testing, i.e. a null hypothesis (e.g., that the data do not contain a signal of interest) is rejected if, under the null, the probability of observing data as extreme or more extreme than what has been observed is small (the so-called p -value).
- An alternative viewpoint is offered by a Bayesian approach to probability, in which the focus is shifted away from rejecting a null hypothesis to comparing alternative explanations. Again as we have seen already, this is done by means of the posterior odds (ratio of probabilities) between two (or several) competing models.
- Nevertheless, without entering in a long-lasting discussion, there is the possibility to compute a *Bayesian p -values* to a model, as discussed in [Lucy \(2016\)](#).

Reference & Material

Material and papers related to the topics discussed in this lecture.

- [Trotta \(2017\)](#) - "[Bayesian Methods in Cosmology](#)"
- [Karamanis \(2023\)](#) - "[Bayesian Computation in Astronomy: Novel methods for parallel and gradient-free inference \(Chapter 1-3\)](#)"

Further Material

Papers for examining more closely some of the discussed topics.

- [E.T. Jaynes](#) - "[Probability Theory: The Logic of Science](#)"
- [A. Gelman et al.](#) - "[Bayesian Data Analysis](#)"

- [J. Vandeplas \(2014\) - "Frequentism and Bayesianism: A Python-driven Primer"](#)
- [L.B. Lucy \(2016\) - "Frequentist tests for Bayesian models"](#)

Credits

Course Flow

	Previous lecture	Next lecture	Course Summary
notebook	Reminder of frequentist statistics	Lecture about spectral analysis	Course Summary
html	Reminder of frequentist statistics	Lecture about spectral analysis	Course Summary

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